

PRIDE

Personal Risk Detection

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The PRIDE (Personal Risk Detection) dataset.

Built from 23 test subjects and a period of data collection of one week each, 24 hours per day. It consists of two sets of data:

- The normal conditions dataset (NCDS).
Comprises the test subject's behavior and physiological patterns in their daily life. Additionally, includes 14 online labelled activities such as having breakfast, having lunch, having dinner, drinking coffee, driving a car, watching TV/going to the cinema, sleeping, riding a bike, jogging, running, a social gathering, going on public transport, and gym (non-aerobic activity).*
- The anomaly conditions dataset (ACDS).
Encompasses data under specific abnormal or stressed conditions. The scenarios include the following activities: rushing 100 meters as fast as possible, going up and down the stairs of a several-floor building as fast as possible, simulating a two-minute fight, simulating falls back and forth (swoons), and simulating breathing problems, by holding out breath as long as possible.

The test subjects group comprises 6 female and 15 male volunteers with ages between 21 and 52 years old, heights from 1.56 m to 1.86 m, weights ranging from 42 to 101 kg, exercising 0 to 10 hours a week, and time spent sitting during working hours or leisure from 20 to 84 hours a week.

A day log from a test subject includes all observations that appear from 00:00 h to 23:59 h. Even though a test subject gathered data for a period of seven days, there is a gap in a log of at least 40 minutes normally three times per day due to the battery recharging process. Moreover, since the collection of data is performed during ordinary and quotidian life, test subjects normally paused data collection for several personal reasons, e.g. to perform an aquatic activity.

To collect data we use Microsoft Band© and a mobile application we developed using the available SDK. The application and PRIDE's raw data are available upon request to ltrejo@itesm.mx.

A pre-processed PRIDE dataset.

As used to obtain result reported in: **Online Personal Risk Detection Based on Behavioural and Physiological Patterns.**

Procedure description:

- For each test subject in the PRIDE dataset, the data collected is arranged by day.

* The 14 labelled activities are only available in the PRIDE raw dataset.

- Date and activity label features are removed.
- The feature vector is computed using a one second window.
- Mean and standard deviation are computed for gyroscope and accelerometer to fit into the window size.
- Absolute values of heart-rate, skin temperature, pace, speed, and UV sensors are preserved.
- The incremental change (Δ -value) for absolute values of total steps, total distance, and calories burned are calculated. The Δ -value is computed as the difference between the current and previous one.

Feature Vector.

A 26 dimensional feature vector is obtained. Following are described the features in order of their appearance in the CSV files.

Feature 1 - Mean of X axis of Gyroscope Accelerometer.

Feature 2 - Standard deviation of X axis of Gyroscope Accelerometer.

Feature 3 - Mean of Y axis of Gyroscope Accelerometer.

Feature 4 - Standard deviation of Y axis of Gyroscope Accelerometer.

Feature 5 - Mean of Z axis of Gyroscope Accelerometer.

Feature 6 - Standard deviation of Z axis of Gyroscope Accelerometer.

Feature 7 - Mean of X axis of Gyroscope Angular Velocity.

Feature 8 - Standard deviation of X axis of Gyroscope Angular Velocity.

Feature 9 - Mean of Y axis of Gyroscope Angular Velocity.

Feature 10 - Standard deviation of Y axis of Gyroscope Angular Velocity.

Feature 11 - Mean of Z axis of Gyroscope Angular Velocity.

Feature 12 - Standard deviation of Z axis of Gyroscope Angular Velocity.

Feature 13 - Mean of X axis of Accelerometer.

Feature 14 - Standard deviation of X axis of Accelerometer.

Feature 15 - Mean of Y axis of Accelerometer.

Feature 16 - Standard deviation of Y axis of Accelerometer.

Feature 17 - Mean of Z axis of Accelerometer.

Feature 18 - Standard deviation of Z axis of Accelerometer.

Feature 19 - Heart-Rate.

Feature 20 - Skin-Temperature.

Feature 21 - Δ Pedometer.

Feature 22 - Δ Distance.

Feature 23 - Speed.

Feature 24 - Pace.

Feature 25 - Δ calories.

Feature 26 - UV.

Dataset files.

23 zip files.

Per zip file:

Two folders:

Dataset folder.

Comprises two archives:

1) ACDS.csv

The anomaly conditions dataset. Encompasses data under specific abnormal or stressed conditions.

2) NCDS.csv

The normal conditions dataset. Comprises the test subject's behavior and physiological patterns in their daily life.

Five Fold Cross Validation folder.

Each of the test subject logs are divided into 5 numbered folds to use them in a five-fold cross-validation.

Each fold holds two archives:

1) training.csv.

This file encompasses 4/5 parts of test subject's normal conditions dataset (NCDS).

2) testing.csv.

This file combines 1/5 part of test subject's normal conditions dataset (NCDS) and all data from anomaly conditions dataset (ACDS) of the same test subject.

A last column of every row contains one of two labels (“typical” or “atypical”). The label typical indicates that the object represents normal test subject behaviour, and the label “atypical” indicates that the object represents a personal risk situation.

Important Notes.

We are very grateful for your interest in the dataset. Additionally, we appreciate if you send an email to ltrejo@itesm.mx to inform us of any publication using this dataset or any variation of it. Thus we can reference your research on this document.

Disclaimer.

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